

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I D.Anandpaul(192425207) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

1. Internet Connectivity for a Rural Branch Office

Introduction

A multinational company is establishing a branch office in a remote rural area where high-speed wired internet is unavailable. The network must support cloud applications, video conferencing, and secure communication with headquarters at a reasonable cost.

Requirements and Constraints

- High-speed internet is required for cloud applications and video conferencing.
- Wired broadband is unavailable.
- The budget is limited.
- Continuous business operation is required.
- Secure communication with headquarters is necessary.
- The network should support future expansion.

Proposed Solution

- Use 4G/5G fixed wireless internet as the primary connection.
- Use satellite internet as a backup connection.
- Install a dual-WAN router for automatic connection failover.
- Use a Gigabit Ethernet switch to connect office devices.
- Deploy Wi-Fi 6 access points for wireless users.
- Use a firewall and VPN for security.
- Use WPA3 for secure Wi-Fi access.

Justification

4G/5G provides good bandwidth without the high installation cost of wired infrastructure. Satellite internet provides backup connectivity when the primary connection fails. A dual-WAN router automatically switches to the backup connection, improving reliability. VPN encryption protects communication with headquarters. The network can later be upgraded to fiber or additional wireless capacity.

Conclusion

The proposed 4G/5G + satellite backup network provides a good balance of bandwidth, reliability, security, cost, and scalability for the rural branch office.

2. Campus-Wide Wireless Network for 3,000+ Students

Introduction

A university needs campus-wide wireless connectivity for more than 3,000 students in classrooms, laboratories, libraries, and hostels. The network must handle many simultaneous users while maintaining security and minimizing interference.

Requirements and Constraints

- More than 3,000 simultaneous users.
- Coverage required across classrooms, labs, libraries, and hostels.
- Limited number of access points due to budget.
- Minimum wireless interference.
- High performance is required.
- Strong user authentication and security are required.

Proposed Solution

- Deploy Wi-Fi 6/6E access points at strategically selected locations.
- Place high-capacity APs in crowded areas such as libraries, hostels, and lecture halls.
- Use 5 GHz and 6 GHz bands for high-speed connections.
- Use 2.4 GHz for devices requiring longer range.
- Use proper channel planning to reduce interference.
- Use WPA3-Enterprise with 802.1X/RADIUS authentication.
- Create separate networks for students, staff, and guests.
- Use centralized wireless management to monitor all access points.

Justification

Strategic AP placement provides maximum coverage using fewer devices, reducing cost. The 5 GHz and 6 GHz bands provide higher speed and more channels, while channel planning reduces interference. WPA3-Enterprise provides secure individual authentication. Separate networks prevent students and guests from accessing sensitive university resources.

Conclusion

A centrally managed Wi-Fi 6/6E network provides an effective balance between coverage, performance, security, interference control, and budget.

3. Secure Network Architecture for a Financial Institution

Introduction

A financial institution handles highly sensitive customer information and online banking services. Therefore, its internal network must provide strong cybersecurity while maintaining good performance and meeting security and compliance requirements.

Requirements and Constraints

- Customer information must be protected.
- Employees need controlled access to sensitive systems.
- Online banking services must remain available.
- Strong security and compliance are required.
- Network performance should not be significantly reduced.
- Security implementation should remain cost-effective.

Proposed Solution

- Divide the network into separate VLANs for employees, servers, guests, and sensitive systems.
- Create a DMZ for public-facing online banking/web servers.
- Deploy a Next-Generation Firewall (NGFW) between external and internal networks.
- Use MFA and 802.1X/RADIUS for employee authentication.
- Implement Role-Based Access Control (RBAC).
- Use IDS/IPS to detect and prevent suspicious activities.
- Use VPN and TLS encryption for secure communication.
- Maintain centralized security logs for monitoring and auditing.

Justification

Network segmentation limits unauthorized access and reduces the impact of a security breach. The DMZ separates public-facing servers from internal systems. The firewall controls network traffic, while IDS/IPS detects malicious activity. MFA provides stronger authentication than passwords alone. These measures provide strong security without unnecessarily affecting normal network performance.

Conclusion

The proposed segmented and layered security architecture provides a strong balance between security, performance, compliance, cost, and scalability, making it suitable for a financial institution.